2 Modules

Exercise 2.1. Show that $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) = 0$ if m, n are coprime.

Proof: We will use the idea of 2.2, makes proof much easier. Consider the exact sequence

$$\mathbb{Z} \overset{\times m}{\to} \mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \to 0.$$

Apply the $\cdot \otimes_A \mathbb{Z}/n\mathbb{Z}$ functor.

$$\mathbb{Z}/n\mathbb{Z} \overset{\times m}{\to} \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

Note that $\times m$ is a isomorphism since m, n are co prime. Hence

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = 0$$

□

Exercise 2.2. Let A be a ring, a an ideal, M an A-module. Show that $(A/a) \otimes_A M$ is isomorphic to M/aM. [Tensor the exact sequence $0 \to a \to A \to A/a \to 0$ with M.]

Proof: Note the following, $a \overset{i}{\hookrightarrow} A \to A/a \to 0$. Tensoring by M preserves exactness,

$$a \otimes_A M \overset{i\otimes 1}{\to} A \otimes_A M \to A/a \otimes_A M \to 0.$$

Thus

$$A/a \otimes_A M \cong (A \otimes_A M)/((i \otimes 1)(a \otimes_A M)) \cong M/aM.$$

The last isomorphism comes from the canonical isomorphism $\varphi : A \otimes_A M \to M$ where $(\alpha, m) \to \alpha m$.

$$\begin{aligned}(\varphi \circ (i \otimes 1))(a \otimes_A M) &= \varphi\left(\left\{\sum_{k=1}^{n} a_k \otimes m_k : n \in \mathbb{N}, a_k \in a, m_k \in M\right\}\right) \\ &= \left\{\sum_{k=1}^{n} a_k m_k : n \in \mathbb{N}, a_k \in a, m_k \in M\right\} = aM\end{aligned}$$

□

Exercise 2.3. Let A be a local ring, M and N finitely generated A-modules. Prove that if $M \otimes N = 0$, then $M = 0$ or $N = 0$. [Let m be the maximal ideal, $k = A/m$ the residue field. Let $M_k = k \otimes_A M \cong M/mM$ by Exercise 2. By Nakayama's lemma, $M_k = 0 \Rightarrow M = 0$. But $M \otimes_A N = 0 \Rightarrow (M \otimes_A N)_k = 0 \Rightarrow M_k \otimes_k N_k = 0 \Rightarrow M_k = 0$ or $N_k = 0$, since M_k, N_k are vector spaces over a field.]

Exercise 2.4. Let M_i $(i \in I)$ be any family of A-modules, and let M be their direct sum. Prove that M is flat $\Leftrightarrow$ each M_i is flat.

Exercise 2.5. Let $A[x]$ be the ring of polynomials in one indeterminate over a ring A. Prove that $A[x]$ is a flat A-algebra. [Use Exercise 4.]

Exercise 2.6. For any A-module M, let $M[x]$ denote the set of all polynomials in x with coefficients in M, that is to say expressions of the form

$$m_0 + m_1 x + ... + m_r x^r \quad (m_i \in M).$$

Defining the product of an element of $A[x]$ and an element of $M[x]$ in the obvious way, show that $M[x]$ is an $A[x]$-module.

Show that $M[x] \cong A[x] \otimes_A M$.

Proof: We will prove this by showing that $M[x]$ satisfies the universal property of $A[x] \otimes_A M$. Let

$$\Theta : A[x] \times M \to M[x]$$
$$\left(\sum_{k=0}^{n} a_k x^k, m\right) \mapsto \sum_{k=0}^{n} a_k m x^k.$$

It is easy to check that Θ is an A-bilinear map. Let (N, φ) be an arbitrary pair of an A-module and an A-bilinear map $(A[x], M) \xrightarrow{\varphi} N$. We will show that there is a unique A-linear map from $\psi : M[x] \to N$ such that $\psi \circ \Theta = \varphi$. Note the following, $S = \{mx^k : m \in M, k \in \mathbb{N}_0\}$ forms a generating set of $M[x]$ as an A-module. Thus by fixing how a map behaves on S and knowing that it is A-linear, it determines its behavior on $M[x]$.

i) Existence: Let $\psi(mx^k) = \varphi(x^k, m)$. Thus:

$$\psi\left(\sum_{k=0}^{n} m_k x^k\right) = \sum_{k=0}^{n} \varphi(x^k, m_k).$$

We will check well definedness, suppose

$$\sum_{k=0}^{n} m_k x^k = \sum_{k=0}^{n} {m_k}' x^k \Rightarrow \sum_{k=0}^{n} (m_k - {m'}_k) x^k = 0 \Rightarrow m_k = {m'}_k.$$

A-linearity is immediate.

$$\begin{aligned}
\psi\left(\Theta\left(\sum_{k=0}^{n} a_k x^k, m\right)\right) &= \psi\left(\sum_{k=0}^{n} a_k m x^k\right) \\
&= \sum_{k=0}^{n} \varphi(x^k, a_k m) \\
&= \sum_{k=0}^{n} \varphi(a_k x^k, m) \\
&= \varphi\left(\sum_{k=0}^{n} a_k x^k, m\right)
\end{aligned}$$

ii) Uniqueness: Suppose ψ and ψ' satisfy the necessary conditions. We know that

$$\psi(mx^k) = \psi(\Theta(x^k, m)) = \varphi(x^k, m) = \psi'(\Theta(x^k, m)) = \psi'(mx^k).$$

Since ψ and ψ' agree on a generating set of $M[x]$, and are both A-linear, they are equal.

Thus $(M[x], \Theta)$ satisfies the universal property of the tensor product of A-modules, and thus $M[x] \cong A[x] \otimes_A M$ as A-modules.

This isomorphism is $A[x]$-linear. The induced map $A[x] \otimes_A M \to M[x]$ sends $x^k \otimes m \mapsto mx^k$, so multiplication by x on either side gives $x^{k+1} \otimes m \mapsto mx^{k+1}$. Since A and x generate $A[x]$ as a ring, the map commutes with the action of all of $A[x]$.

Alternate Proof: Let N be an arbitrary $A[x]$-module. We will show that there are bijections, natural in N,

$$\mathrm{Hom}_{A[x]}(A[x] \otimes_A M, N) \leftrightarrow \mathrm{Hom}_A(M, N) \leftrightarrow \mathrm{Hom}_{A[x]}(M[x], N).$$

(Here $M[x]$ is an $A[x]$-module by part 1.) Naturality is clear: for $h : N \to N'$ an $A[x]$-linear map, $(h \circ \varphi) \circ i = h \circ (\varphi \circ i)$, and similarly for i_0 below. Hence the functors $\mathrm{Hom}_{A[x]}(A[x] \otimes_A M, -)$ and $\mathrm{Hom}_{A[x]}(M[x], -)$ are naturally isomorphic, and by Yoneda $M[x] \cong A[x] \otimes_A M$ as $A[x]$-modules.

i) Let $i : M \to A[x] \otimes_A M$ where $m \mapsto 1 \otimes m$. Define

$$T : \mathrm{Hom}_{A[x]}(A[x] \otimes_A M, N) \to \mathrm{Hom}_A(M, N)$$
$$\varphi \mapsto \varphi \circ i.$$

An $A[x]$-linear φ is automatically A-linear, and it corresponds to an A-bilinear map $\hat{\varphi} : A[x] \times M \to N$ where

$$\begin{aligned} \hat{\varphi}(f, m) &= \varphi(f \otimes m) \\ &= f\varphi(1 \otimes m) \\ &= f(\varphi \circ i)(m). \end{aligned}$$

Thus φ is determined by $\varphi \circ i$, so T is injective. For surjectivity, let $\psi \in \mathrm{Hom}_A(M, N)$. The map $\hat{\psi} : A[x] \times M \to N$, $(f, m) \mapsto f\psi(m)$, is A-bilinear, so it induces an A-linear $\varphi : A[x] \otimes_A M \to N$ with $\varphi(f \otimes m) = f\psi(m)$. It is $A[x]$-linear: $\varphi(g(f \otimes m)) = gf\psi(m) = g\varphi(f \otimes m)$ for all $g \in A[x]$. Also $T(\varphi) = \varphi \circ i = \psi$. Thus T is a bijection.

ii) Let $i_0 : M \to M[x]$ where $m \mapsto m$ (that is, mx^0). Define

$$S : \mathrm{Hom}_{A[x]}(M[x], N) \to \mathrm{Hom}_A(M, N)$$
$$\gamma \mapsto \gamma \circ i_0.$$

Injectivity: Let $\gamma \in \mathrm{Hom}_{A[x]}(M[x], N)$ and $u = \gamma \circ i_0$. Then $\gamma(mx^k) = x^k\gamma(mx^0) = x^k u(m)$. Any two maps with the same u agree on the generators mx^k, hence by $A[x]$-linearity agree everywhere. Surjectivity: Let $u \in \mathrm{Hom}_A(M, N)$ and define

$$\gamma\left(\sum_{k=0}^{n} m_k x^k\right) = \sum_{k=0}^{n} x^k u(m_k).$$

This is additive and A-linear because u is, and it commutes with multiplication by x. Since A and x generate $A[x]$ as a ring, γ is $A[x]$-linear. Also $\gamma \circ i_0 = u$. Thus S is a bijection.

Thus T and S are bijections, natural in N, and the claim follows by Yoneda as above. □

Exercise 2.7. Let p be a prime ideal in A. Show that $p[x]$ is a prime ideal in $A[x]$. If m is a maximal ideal in A, is $m[x]$ a maximal ideal in $A[x]$?

Proof: **Direct Argument**: Consider $\bmod p : A[x] \to (A/p)[x]$ where

$$\sum_{k=1}^{n} a_k x^k \to \sum_{k=1}^{n} (a_k \bmod p) x^k.$$

$\bmod p$ is obvously surjective. If $k \in \ker(\bmod p) \Rightarrow$ all coefficients of k are in p, hence $k \in p[x]$. Use First Isomorphism theorem. Finally polynomials over a domain is a domain. No, this doesn't work for maximal ideals: consider $\mathbb{Z}[x]$ and $(2\mathbb{Z})[x]$, $\mathbb{F}_2[x]$ is not a domain. . □

Exercise 2.8.
i) If M and N are flat A-modules, then so is $M \otimes_A N$.
ii) If B is a flat A-algebra and N is a flat B-module, then N is flat as an A-module.

Exercise 2.9. Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of A-modules. If M' and M'' are finitely generated, then so is M.

Exercise 2.10. Let A be a ring, a an ideal contained in the Jacobson radical of A; let M be an A-module and N a finitely generated A-module, and let $u : M \to N$ be a homomorphism. If the induced homomorphism $M/aM \to N/aN$ is surjective, then u is surjective.

Exercise 2.11. Let A be a ring $\neq 0$. Show that $A^m \cong A^n \Rightarrow m = n$. [Let m be a maximal ideal of A and let $\varphi : A^m \to A^n$ be an isomorphism. Then $1 \otimes \varphi : (A/m) \otimes_A A^m \to (A/m) \otimes_A A^n$ is an isomorphism between vector spaces of dimensions m and n over the field $k = A/m$. Hence $m = n$.] (Cf. Chapter 3, Exercise 15.)

If $\varphi : A^m \to A^n$ is surjective, then $m \geq n$.

If $\varphi : A^m \to A^n$ is injective, is it always the case that $m \leq n$?

Exercise 2.12. Let M be a finitely generated A-module and $\varphi : M \to A^n$ a surjective homomorphism. Show that $\ker(\varphi)$ is finitely generated. [Let $e_1, ..., e_n$ be a basis of A^n and choose $u_i \in M$ such that $\varphi(u_i) = e_i$ $(1 \leq i \leq n)$. Show that M is the direct sum of $\ker(\varphi)$ and the submodule generated by $u_1, ..., u_n$.]

Exercise 2.13. Let $f : A \to B$ be a ring homomorphism, and let N be a B-module. Regarding N as an A-module by restriction of scalars, form the B-module $N_B = B \otimes_A N$. Show that the homomorphism $g : N \to N_B$ which maps y to $1 \otimes y$ is injective and that $g(N)$ is a direct summand of N_B. [Define $p : N_B \to N$ by $p(b \otimes y) = by$, and show that $N_B = \mathrm{im}(g) \oplus \ker(p)$.]

2.1 Direct limits

Exercise 2.14. A partially ordered set I is said to be a **directed set** if for each pair i, j in I there exists $k \in I$ such that $i \leq k$ and $j \leq k$.

Let A be a ring, let I be a directed set and let $(M_i)_{i \in I}$ be a family of A-modules indexed by I. For each pair i, j in I such that $i \leq j$, let $\mu_{ij} : M_i \to M_j$ be an A-homomorphism, and suppose that the following axioms are satisfied: (1) μ_{ii} is the identity mapping of M_i, for all $i \in I$; (2) $\mu_{ik} = \mu_{jk} \circ \mu_{ij}$ whenever $i \leq j \leq k$.

Then the modules M_i and homomorphisms μ_{ij} are said to form a **direct system** $\boldsymbol{M} = (M_i, \mu_{ij})$ over the directed set I.

We shall construct an A-module M called the **direct limit** of the direct system $\boldsymbol{M}$. Let C be the direct sum of the M_i, and identify each module M_i with its canonical image in C. Let D be the submodule of C generated by all elements of the form $x_i - \mu_{ij}(x_i)$ where $i \leq j$ and $x_i \in M_i$. Let $M = C/D$, let $\mu : C \to M$ be the projection and let μ_i be the restriction of μ to M_i.

The module M, or more correctly the pair consisting of M and the family of homomorphisms $\mu_i : M_i \to M$, is called the **direct limit** of the direct system $\boldsymbol{M}$, and is written $\lim_{\to} M_i$. From the construction it is clear that $\mu_i = \mu_j \circ \mu_{ij}$ whenever $i \leq j$.

Exercise 2.15. In the situation of Exercise 14, show that every element of M can be written in the form $\mu_i(x_i)$ for some $i \in I$ and some $x_i \in M_i$.

Show also that if $\mu_i(x_i) = 0$ then there exists $j \geq i$ such that $\mu_{ij}(x_i) = 0$ in M_j.

Exercise 2.16. Show that the direct limit is characterized (up to isomorphism) by the following property. Let N be an A-module and for each $i \in I$ let $\alpha_i : M_i \to N$ be an A-module homomorphism such that $\alpha_i = \alpha_j \circ \mu_{ij}$ whenever $i \leq j$. Then there exists a unique homomorphism $\alpha : M \to N$ such that $\alpha_i = \alpha \circ \mu_i$ for all $i \in I$.

Exercise 2.17. Let $(M_i)_{i \in I}$ be a family of submodules of an A-module, such that for each pair of indices i, j in I there exists $k \in I$ such that $M_i + M_j \subseteq M_k$. Define $i \leq j$ to mean $M_i \subseteq M_j$ and let $\mu_{ij} : M_i \to M_j$ be the embedding of M_i in M_j. Show that

$$\lim_{\to} M_i = \sum_i M_i = \cup_i M_i.$$

In particular, any A-module is the direct limit of its finitely generated submodules.

Exercise 2.18. Let $\boldsymbol{M} = \big(M_i, \mu_{ij}\big)$, $\boldsymbol{N} = \big(N_i, \nu_{ij}\big)$ be direct systems of A-modules over the same directed set. Let M, N be the direct limits and $\mu_i : M_i \to M$, $\nu_i : N_i \to N$ the associated homomorphisms.

A **homomorphism** $\Phi : \boldsymbol{M} \to \boldsymbol{N}$ is by definition a family of A-module homomorphisms $\varphi_i : M_i \to N_i$ such that $\varphi_j \circ \mu_{ij} = \nu_{ij} \circ \varphi_i$ whenever $i \leq j$. Show that Φ defines a unique homomorphism $\varphi = \lim_{\to} \varphi_i : M \to N$ such that $\varphi \circ \mu_i = \nu_i \circ \varphi_i$ for all $i \in I$.

Exercise 2.19. A sequence of direct systems and homomorphisms

$$\boldsymbol{M} \to \boldsymbol{N} \to \boldsymbol{P}$$

is **exact** if the corresponding sequence of modules and module homomorphisms is exact for each $i \in I$. Show that the sequence $\lim_{\to} M_i \to \lim_{\to} N_i \to \lim_{\to} P_i$ of direct limits is then exact. [Use Exercise 15.]

2.2 Tensor products commute with direct limits

Exercise 2.20. Keeping the same notation as in Exercise 14, let N be any A-module. Then $\big(M_i \otimes N, \mu_{ij} \otimes 1\big)$ is a direct system; let $P = \lim_{\to}(M_i \otimes N)$ be its direct limit.

For each $i \in I$ we have a homomorphism $\mu_i \otimes 1 : M_i \otimes N \to M \otimes N$, hence by Exercise 16 a homomorphism $\psi : P \to M \otimes N$. Show that ψ is an isomorphism, so that

$$\varinjlim(M_i \otimes N) \cong \left(\varinjlim M_i\right) \otimes N.$$

[For each $i \in I$, let $g_i : M_i \times N \to M_i \otimes N$ be the canonical bilinear mapping. Passing to the limit we obtain a mapping $g : M \times N \to P$. Show that g is A-bilinear and hence define a homomorphism $\varphi : M \otimes N \to P$. Verify that $\varphi \circ \psi$ and $\psi \circ \varphi$ are identity mappings.]

Exercise 2.21. Let $(A_i)_{i \in I}$ be a family of rings indexed by a directed set I, and for each pair $i \leq j$ in I let $\alpha_{ij} : A_i \to A_j$ be a ring homomorphism, satisfying conditions (1) and (2) of Exercise 14. Regarding each A_i as a $\mathbb{Z}$-module we can then form the direct limit $A = \lim_{\to} A_i$. Show that A inherits a ring structure from the A_i so that the mappings $A_i \to A$ are ring homomorphisms. The ring A is the **direct limit** of the system (A_i, α_{ij}).

If $A = 0$ prove that $A_i = 0$ for some $i \in I$. [Remember that all rings have identity elements!]

Exercise 2.22. Let (A_i, α_{ij}) be a direct system of rings and let $\mathfrak{n}_i$ be the nilradical of A_i. Show that $\lim_{\to} \mathfrak{n}_i$ is the nilradical of $\lim_{\to} A_i$.

If each A_i is an integral domain, then $\lim_{\to} A_i$ is an integral domain.

Exercise 2.23. Let $(B_\lambda)_{\lambda \in \Lambda}$ be a family of A-algebras. For each finite subset J of Λ let B_J denote the tensor product (over A) of the B_λ for $\lambda \in J$. If J' is another finite subset of Λ, and $J \subseteq J'$, there is a canonical A-algebra homomorphism $B_J \to B_{J'}$. Let B denote the direct limit of the rings B_J as J runs through all finite subsets of Λ. The ring B has a natural A-algebra structure for which the homomorphisms $B_J \to B$ are A-algebra homomorphisms. The A-algebra B is the **tensor product** of the family $(B_\lambda)_{\lambda \in \Lambda}$.

2.3 Flatness and Tor

In these exercises it will be assumed that the reader is familiar with the definition and basic properties of the Tor functor.

Exercise 2.24. If M is an A-module, the following are equivalent:

i) M is flat;
ii) $\operatorname{Tor}_n^A(M, N) = 0$ for all $n > 0$ and all A-modules N;
iii) $\operatorname{Tor}_1^A(M, N) = 0$ for all A-modules N.

[To show that (i) $\Rightarrow$ (ii), take a free resolution of N and tensor it with M. Since M is flat, the resulting sequence is exact and therefore its homology groups, which are the $\operatorname{Tor}_n^A(M, N)$, are zero for $n > 0$. To show that (iii) $\Rightarrow$ (i), let $0 \to N' \to N \to N'' \to 0$ be an exact sequence. Then, from the Tor exact sequence,

$$\operatorname{Tor}_1(M, N'') \to M \otimes N' \to M \otimes N \to M \otimes N'' \to 0$$

is exact. Since $\operatorname{Tor}_1(M, N'') = 0$ it follows that M is flat.]

Exercise 2.25. Let $0 \to N' \to N \to N'' \to 0$ be an exact sequence, with N'' flat. Then N' is flat $\Leftrightarrow$ N is flat. [Use Exercise 24 and the Tor exact sequence.]

Exercise 2.26. Let N be an A-module. Then N is flat $\Leftrightarrow$ $\operatorname{Tor}_1(A/a, N) = 0$ for all finitely generated ideals a in A.

[Show first that N is flat if $\mathrm{Tor}_1(M,N)=0$ for all finitely generated A-modules M, by using (2.19). If M is finitely generated, let $x_1,...,x_n$ be a set of generators of M, and let M_i be the submodule generated by $x_1,...,x_i$. By considering the successive quotients M_i/M_{i-1} and using Exercise 25, deduce that N is flat if $\mathrm{Tor}_1(M,N)=0$ for all **cyclic** A-modules M, i.e., all M generated by a single element, and therefore of the form A/a for some ideal a. Finally use (2.19) again to reduce to the case where a is a finitely generated ideal.]

Exercise 2.27. A ring A is **absolutely flat** if every A-module is flat. Prove that the following are equivalent:

i) A is absolutely flat.
ii) Every principal ideal is idempotent.
iii) Every finitely generated ideal is a direct summand of A.

[i) $\Rightarrow$ ii). Let $x\in A$. Then $A/(x)$ is a flat A-module, hence in the diagram

$$\begin{array}{ccc} (x)\otimes A & \overset{\beta}{\to} & (x)\otimes(A/(x)) \\ \downarrow & & \overset{\alpha}{\downarrow} \\ A & \to & A/(x) \end{array}$$

the mapping α is injective. Hence $\mathrm{im}(\beta)=0$, hence $(x)=(x^2)$. ii) $\Rightarrow$ iii). Let $x\in A$. Then $x=ax^2$ for some $a\in A$, hence $e=ax$ is idempotent and we have $(e)=(x)$. Now if e,f are idempotents, then $(e,f)=(e+f-ef)$. Hence every finitely generated ideal is principal, and generated by an idempotent e, hence is a direct summand because $A=(e)\oplus(1-e)$. iii) $\Rightarrow$ i). Use the criterion of Exercise 26.]

Exercise 2.28. A Boolean ring is absolutely flat. The ring of Chapter 1, Exercise 7 is absolutely flat. Every homomorphic image of an absolutely flat ring is absolutely flat. If a local ring is absolutely flat, then it is a field.

If A is absolutely flat, every non-unit in A is a zero-divisor.